MILITARY SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY

© TYPE 10WP7A 1/

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Magnetic deflection and electrostatic focus, aluminized screen

PIN CONNECTIONS AND DIMENSIONS: See figure 1

ABSOLUTE RATINGS:

Parameter:	Ef	Ec1	Ec2	Eb1	Eb2	Rg	Ehk	Alt
Unit:	V	Vdc	Vdc	Vdc	Vdc	Meg	Vdc	ft
Maximum:	6.9	0, -200	600	900	13,200	Note 1	±180	80,000
Minimum	5.7	---	---	-450	7,000	---	---	---
TEST CONDITIONS:	6.3	Adjust	300	Focus	10,000	---	---	---

GENERAL:

Qualification - Required

1/ Replaces tube types 10WP7 and MIS 12750

© denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Qualification inspection						
1216	Base material insulating quality	-	Zone 5 (min)	---	---	---	---
5216	Cathode illumination	-		---	---	---	---
1002	Barometric pressure, reduced	-	21 mmHg	---	---	---	---
1141	Pressure (implosion)	-		---	---	---	---
1331	Direct-interelectrode capacitance	-	k to all g1 to all	Ck Cg1	--- ---	8.0 10.0	pF pF
5111	Vibration (cathode-ray tubes)	-		Width	---	2.0	mm
5115	Shock	-		---	---	---	---
	Quality conformance inspection, part 1						
5106	Bulb, screen, and faceplate quality	-		---	---	---	---
5201	Voltage breakdown	-		---	---	---	---
5201	Voltage breakdown (magnetic deflection)	-		---	---	---	---
5223	Modulation	-	Ib2 = 100 μAdc	ΔEc1	---	32	Vdc
5231	Spot position (magnetic deflection)	-		---	---	0.5	in.
5101	Face tilt	-		---	---	---	---
5241	Grid cutoff voltage	-		Ec1	-33	-77	Vdc
5251	Grid No. 1 leakage current	-		---	---	---	---
5251	Anode No. 1 leakage current	-		---	---	---	---
5236	Zero-bias anode current (magnetic deflection)	-		---	---	---	---
5206	Gas ratio	-		Gr	---	0.25	---
5101	Neck straightness	2		---	---	---	---
---	Focusing electrode voltage	3	Ib2 = 100 μAdc	Ec1	0	600	Vdc
522b	Line width A (magnetic deflection)	3	Ib2 = 100 μAdc	Width	---	0.55	mm

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 2						
1101	Secureness of base, cap, or insert	-		---	---	---	---
1301	Heater current	-		If	540	660	mA
5251	Heater-cathode leakage current	-		---	---	---	---
5221	Screens	-		---	---	---	---
5216	Stray emission	-	Eb2 = 13,200 Vdc; Ec2 = 600 Vdc; Eb1 = 0	---	---	---	---
5201	Electrode currents (grid No. 2)	-	Ec1 = 0	Ic2	-15	+15	μAdc
5251	Grid No. 2 leakage current	-		---	---	---	---
5101	Neck and bulb alinement	-		---	---	---	---
5101	Side terminal and base alinement	-	Pin No. 3	---	---	---	---
1111	Base pin solder depth	-		---	---	---	---
1105	Permanence of marking	-		---	---	---	---
	Quality conformance inspection, part 3						
---	Life test	-	Group C: t = 500 hours (min); Eb2 = 13,200 Vdc; Ec2 = 600 Vdc; Eb1 = focus; Ib2 = 60 μAdc	---	---	---	---
---	Life-test end points:						
5223	Modulation	-	Ib2 = 100 μAdc	ΔEc1	---	32	Vdc
5226	Line width A	-	Ib2 = 100 μAdc	Width	---	0.55	mm
5251	Heater-cathode leakage current	-		---	---	---	---
5251	Grid No. 1 leakage current	-		---	---	---	---
5251	Grid No. 2 leakage current	-		---	---	---	---
5251	Anode No. 1 leakage current	-		---	---	---	---
5216	Stray light emission (conventional types)	-		---	---	---	---

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NOTES:

1. When Ec2 is greater than 330 Vdc, Rg1 shall not exceed 0.5 Meg. When Ec2 is less than 330 Vdc, Rg1 shall not exceed 1.5 Meg.

2. The neck and base straightness shall be determined by the insertion of the tube neck in a cylinder 5-inches (127.0 mm) long and $1.500 \begin{smallmatrix}+0.003\\-0.000\end{smallmatrix}$ inches ($38.10 \begin{smallmatrix}+0.08\\-0.00\end{smallmatrix}$ mm) inside diameter. This cylinder should move freely between the reference line and the base of the assembled tube.

3. The raster shall overscan the tube face. The centerline of the raster shall be adjusted for best focus.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - MI
Air Force - 17, 80
DSA - ES

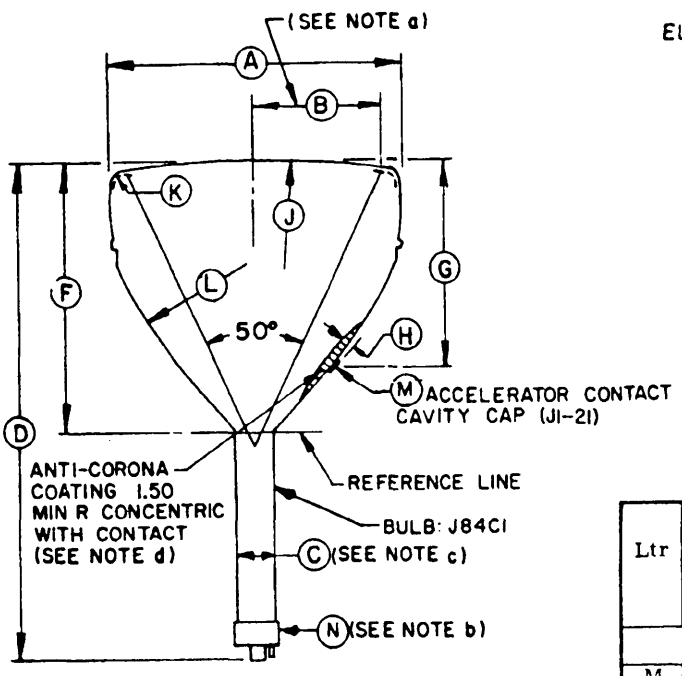
User activities:
Army - WC
Navy - AS, OS, MC, CG, SH
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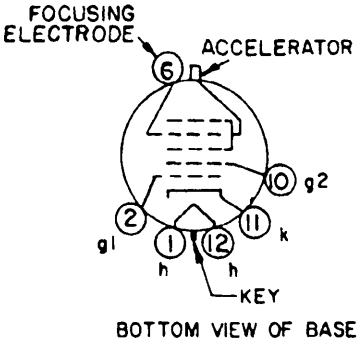
Preparing activity: Navy - EC

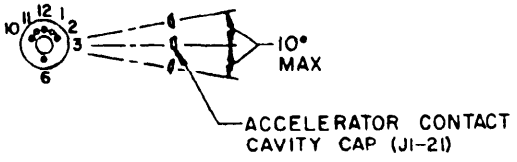
Agent: DSA - ES

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Qualification inspection		
M	Bulb contact: J1-21	
N	Base: B6-63	
Quality conformance inspection, part 2		
A	10.38(263.65)	10.62(269.75)
B	4.50(114.30)	
C	1.38 (35.05)	1.50 (38.10)
D	16.56(420.62)	17.31(439.67)
F	9.25(234.95)	9.62(244.35)
G	6.88(174.75)	7.62(193.55)
H		.19 (4.83)
Reference dimensions (see note e)		
J	42.00 (1066.80)	
K	.38 (9.65)	
L	20.00 (508.00)	

NOTES:
a. The minimum useful screen radius.
b. The base shall be a small shell duodecal 6-pin, B6-63.
c. Dimension C applies to entire length of neck below the reference line. The reference line is determined by the point where a gage 1.500+.003(38.10-.08 mm) inside diameter and 2 inches in length will stop against the bulb body.
d. Do not handle tube by the part of the bulb having the anti-corona coating.
e. Reference dimensions are for information only and are not required for inspection purposes.

© FIGURE 1. Outline drawing of electron tube type 10WP7A.

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